

GAMM analyses

2026-04-20

Data preparations

In GAMM analyses, several key variables are required. Primarily, the time-series analysis utilizes Y and X derived from the `looking_time` and `trial_number` variables. A new measure, `looking_time2`, was created to incorporate looking time during the freeze-frame period.

The primary focus of this analysis is the experimental conditions. For this MBE, these are nested within `trial_type` and have been divided into three new variables: infant-directedness (`tt_dir`), consistency (`tt_con`), and social context (`tt_soc`). Any null values have been recoded as dummy values.

Interaction variables include `method` and `participant_sex`. To account for developmental trends, we can utilize either the continuous variable `age_days` or the discrete variable `age_group_new`. While `stimulus_id` could be recoded to further investigate specific stimulus types, this is likely unnecessary for the current analysis.

The random effect is accounted for by the `unique_subject_identifier` variable.

Exclusions:

- `stimulus_type` was excluded as it contains only one unique value.
- `project_id` and `lab_id` were excluded because they represent higher-level groupings than `trial_type` and `unique_subject_identifier`, respectively.
- Demographic variables are not used, since they are controlled altogether by `unique_subject_identifier`, unless we need to look into these details.

Overview

```
names(df)
```

```
## [1] "project_id"      "lab_id"
## [3] "unique_subject_identifier" "method"
## [5] "age_days"        "age_group_new"
## [7] "participant_sex"  "stimulus_type"
## [9] "trial_type"      "trial_number"
## [11] "stimulus_id"     "habituation"
## [13] "looking_time"    "household_size"
## [15] "parent1_education" "parent2_education"
## [17] "participant_place_of_birth" "race"
## [19] "primarycaregiver_gender" "language_group"
## [21] "primary_language" "primary_language_exposure"
## [23] "caregiver_seat"   "second_session"
## [25] "lookingtime_freezeframe"
```

Y: `looking_time` (numeric)

A supplemental variable, `looking_time2`, was created by incorporating looking time from the MB4 freeze-frame period into the conventional looking time measure.

```
df$looking_time2=0
df$looking_time2[!is.na(df$lookingtime_freezeframe)]=df$lookingtime_freezeframe[!is.na(df$lookingtime_freezeframe)]
df$looking_time2=df$looking_time+df$looking_time2
```

X: `trial_number` (numeric)

```
## [1] "MB1: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12"
```

```
## [1] "MB3: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12"
```

```
## [1] "MB4: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12"
```

Exp: `trial_type` (ordered, with dummy)

The `trial_type` column contains nested conditions for the MB1, MB3, and MB5 sub-tasks, which required recategorization for analysis. For instance, to examine the infant-directedness aspect of MB1, a new variable, `tt_dir`, was created to distinguish between 'IDS' (Infant-Directed Speech) and 'ADS' (Adult-Directed Speech). Observations from MB3 and MB5, which did not involve these conditions, were assigned NA values.

Exclusions: The higher-level variable `project_id` was no longer required.

```
unique(df$trial_type)
```

```
## [1] "IDS"          "ADS"          "consistent"   "inconsistent" "nonsocial"
## [6] "social"
```

To facilitate contrast analyses with GAMM, these categories were converted into ordered variables where NA is coded as 0, the target conditions (e.g., IDS/consistent/social) as 1, and the alternative conditions (e.g., ADS/inconsistent/nonsocial) as 2.

```
unique(df$tt_dir) # NA = 0, IDS = 1, ADS = 2
```

```
## [1] 1 2 0
## Levels: 0 < 1 < 2
```

```
unique(df$tt_con) # NA = 0, consistent = 1, inconsistent = 2
```

```
## [1] 0 1 2
## Levels: 0 < 1 < 2
```

```
unique(df$tt_soc) # NA = 0, social = 1, nonsocial = 2
```

```
## [1] 0 2 1  
## Levels: 0 < 1 < 2
```

Additional Exp:

```
# As a "by" factor:  
df$method <- as.factor(df$unique_subject_identifier)  
df$participant_sex <- as.factor(df$participant_sex)
```

```
# Exclusion: only if we would like to look into detailed stimulus types:  
df$stimulus_id <- as.factor(df$stimulus_id)
```

```
# Developmental factors:  
df$age_days <- as.numeric(df$age_days)  
df$age_group_new <- as.factor(df$age_group_new)
```

Random Var: unique_subject_identifier (factor)

Exclusions: The higher-level variable `lab_id` was no longer required.

```
df$unique_subject_identifier <- as.factor(df$unique_subject_identifier)
```

GAMM

```
# Subset data for GAMM analyses
sub_df <- df[, c(
  # Y
  "looking_time",
  "looking_time2",
  # X
  "trial_number",
  # Exp
  "tt_dir", # MB1
  "tt_con", # MB3
  "tt_soc", # MB4
  # By
  "method",
  "participant_sex",
  # Dev
  "age_days",
  "age_group_new",
  #random
  "unique_subject_identifier"
)]

# Sort data per individual trajectory
sub_df = sub_df[order(sub_df$unique_subject_identifier,sub_df$trial_number),]

# Mark the start of each trajectory (for ACF purposes)
sub_df$start.event <- sub_df$trial_number == 1

# Rename variables
names(sub_df) = c("y1","y2","x","dir","con","soc","method","sex","days","agegrp","id","start.event")
```

Still need hime hpc to run *echt* GAMM, but for the time being, use simple random effects

```
s(id, bs = "re") .
```

Autocorrelation not yet corrected!

Overall habituation trend

```

setwd("/Users/yangji/Downloads/osfstorage-archive")
# m0 <- bam(y1 ~ s(x) + s(id, bs = "re"),
#           data = sub_df,
#           method = "fREML",
#           discrete = TRUE)
# saveRDS(m0, file="m0.rds")

m0 = readRDS("m0.rds")

par(mfrow=c(1,2), cex=1.1)

plot_smooth(m0, view="x",
             xlim = c(1,12),
             xlab="Trial",ylab="LT",
             rm.ranef=FALSE, main="intercept+s(Trial)+re", rug=TRUE,
             col='gray')

```

```

## Summary:
## * x : numeric predictor; with 30 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01.

```

```

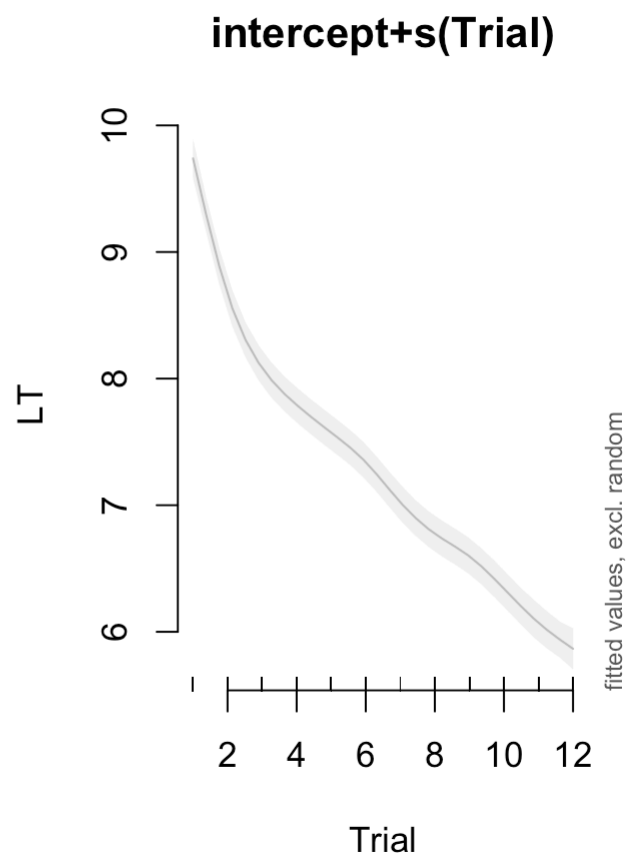
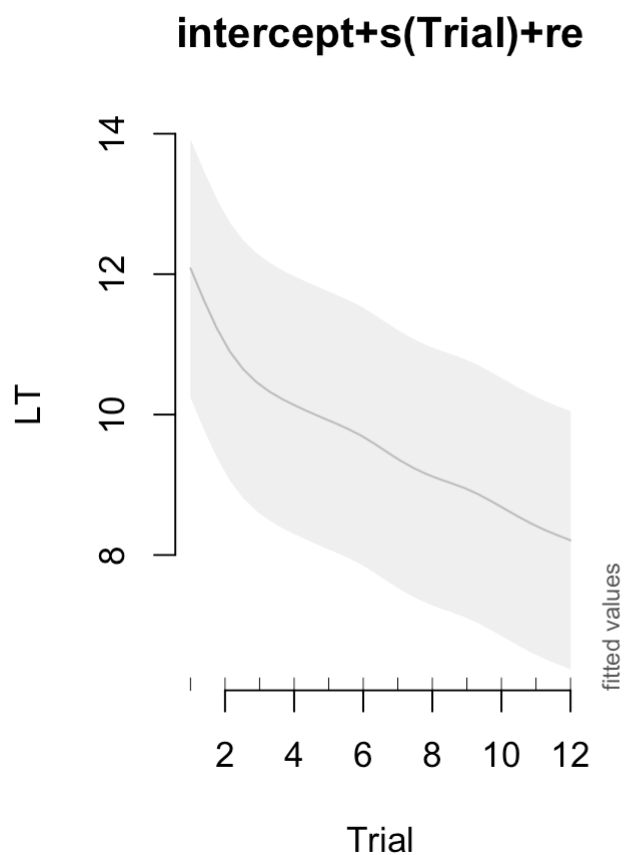
plot_smooth(m0, view="x",
             xlim = c(1,12),
             xlab="Trial",ylab="LT",
             rm.ranef=TRUE, main="intercept+s(Trial)", rug=TRUE,
             col='gray')

```

```

## Summary:
## * x : numeric predictor; with 30 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##

```



```
# contrasts(sub_df$dir) <- 'contr.treatment'
# m1.1 <- bam(y1 ~ s(x) + dir + s(x, by = dir) + s(id, bs = "re"),
#           data = sub_df,
#           method = "fREML",
#           discrete = TRUE)
# saveRDS(m1.1, file="m1.1.rds")

m1.1 = readRDS("m1.1.rds")

summary(m1.1)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## y1 ~ s(x) + dir + s(x, by = dir) + s(id, bs = "re")
##
## Parametric coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.21914    0.09033  90.991  <2e-16 ***
## dir1        -0.97226    0.11724  -8.293  <2e-16 ***
## dir2        -1.88594    0.11724 -16.086  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##           edf   Ref.df      F p-value
## s(x)         1.001     1.001 405.96  <2e-16 ***
## s(x):dir1     7.354     8.330  66.60  <2e-16 ***
## s(x):dir2     7.097     8.135  72.22  <2e-16 ***
## s(id)        3481.468 3854.000  10.27  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.531   Deviance explained = 57.1%
## fREML = 1.1195e+05  Scale est. = 11.016    n = 40961
```

```
# par(mfrow=c(2,2), cex=1.1)
```

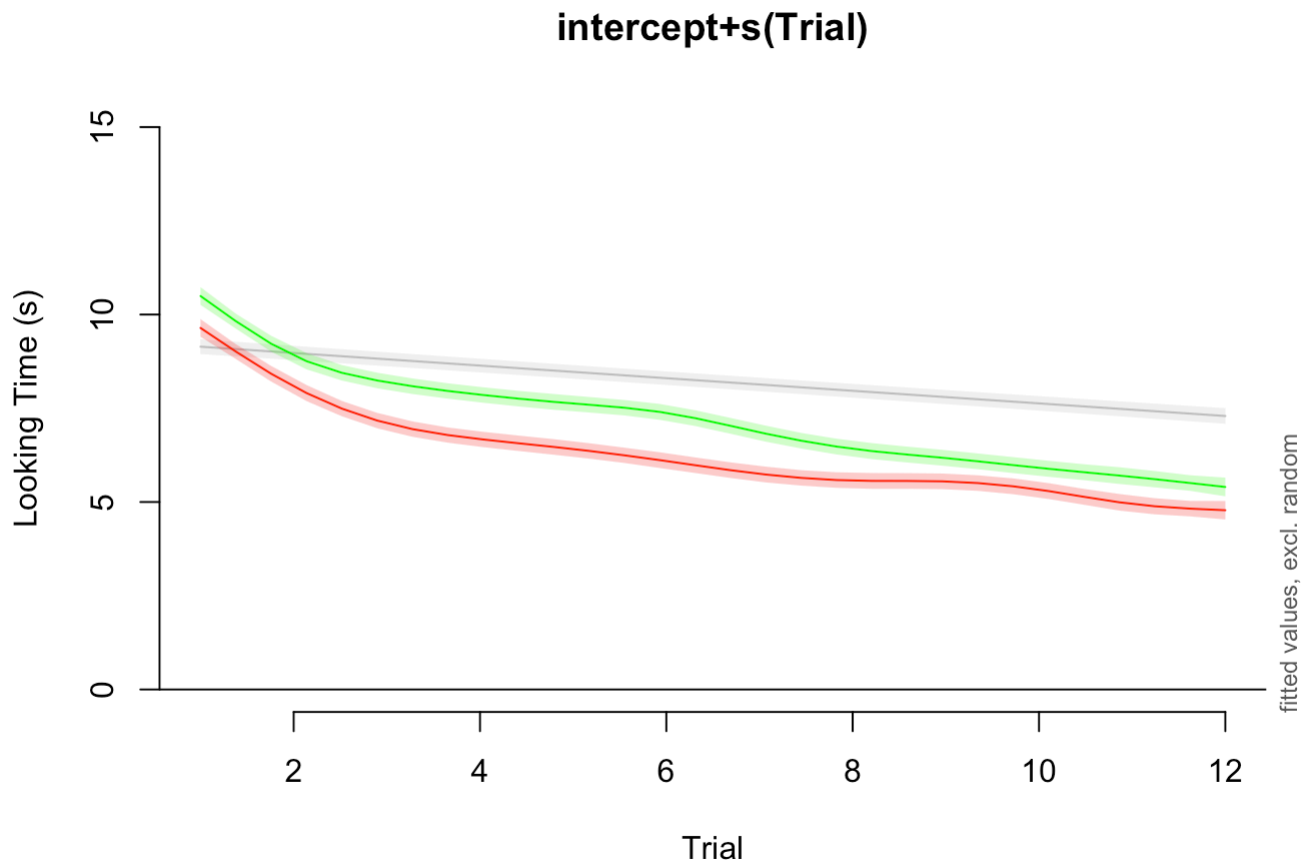
```
plot_smooth(m1.1, view="x", cond=list(dir=0),
             xlim = c(1,12), ylim = c(0,15),
             xlab="Trial", ylab="Looking Time (s)",
             rm.ranef=TRUE, main="intercept+s(Trial)",
             col='gray') ## 0 = NA
```

```
## Summary:
## * dir : numeric predictor; set to the value(s): 0.
## * x : numeric predictor; with 30 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```

```
plot_smooth(m1.1, view="x", cond=list(dir=1),
             add=TRUE, col='green') ## 1 = IDS
```

```
## Summary:
## * dir : numeric predictor; set to the value(s): 1.
## * x : numeric predictor; with 30 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```

```
plot_smooth(m1.1, view="x", cond=list(dir=2),
            add=TRUE, col='red') ## 2 = ADS
```



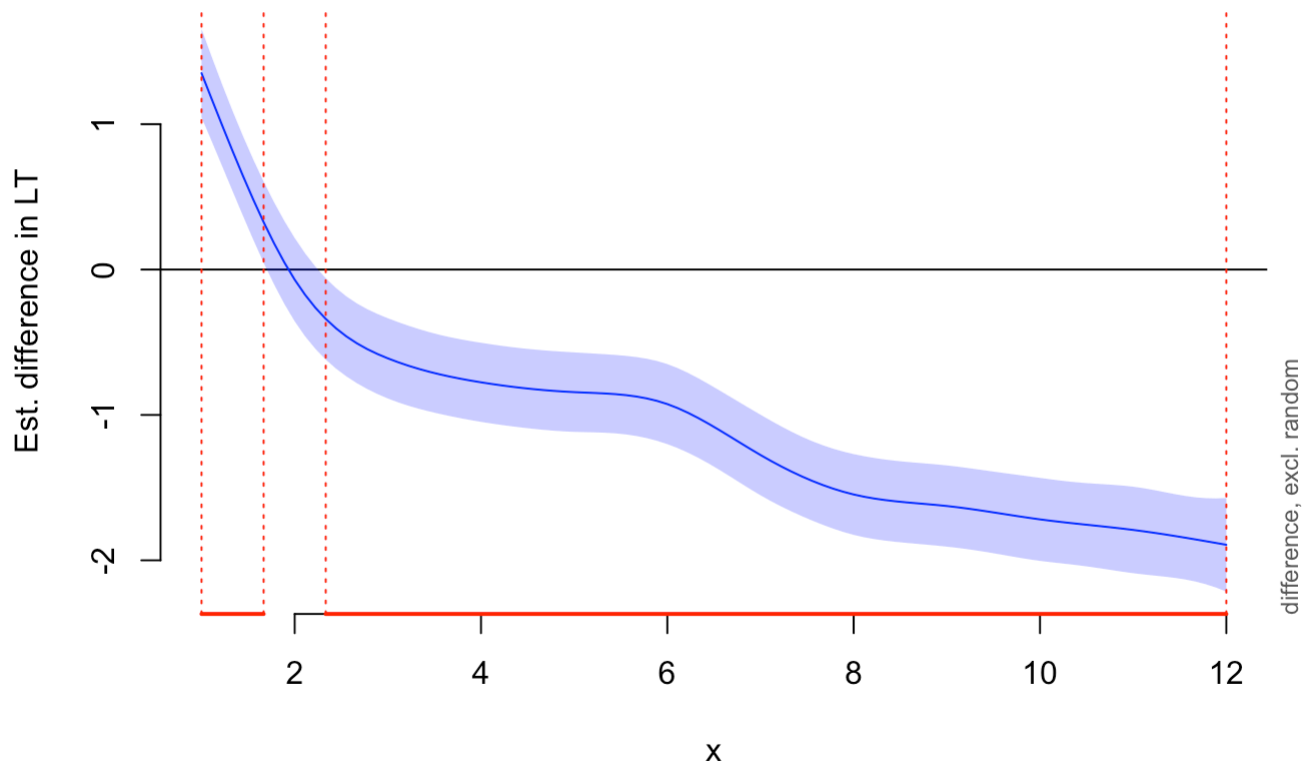
```
## Summary:
## * dir : numeric predictor; set to the value(s): 2.
## * x : numeric predictor; with 30 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```

```
plot_diff(m1.1, view="x",
          comp=list(dir=c(1, 0)),
          # ylim = c(0,8),
          main="IDS-NA",
          col=c('blue','gray'),
          ylab="Est. difference in LT",
          rm.ranef=TRUE)
```



```
## Summary:
## * x : numeric predictor; with 100 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```

IDS-NA

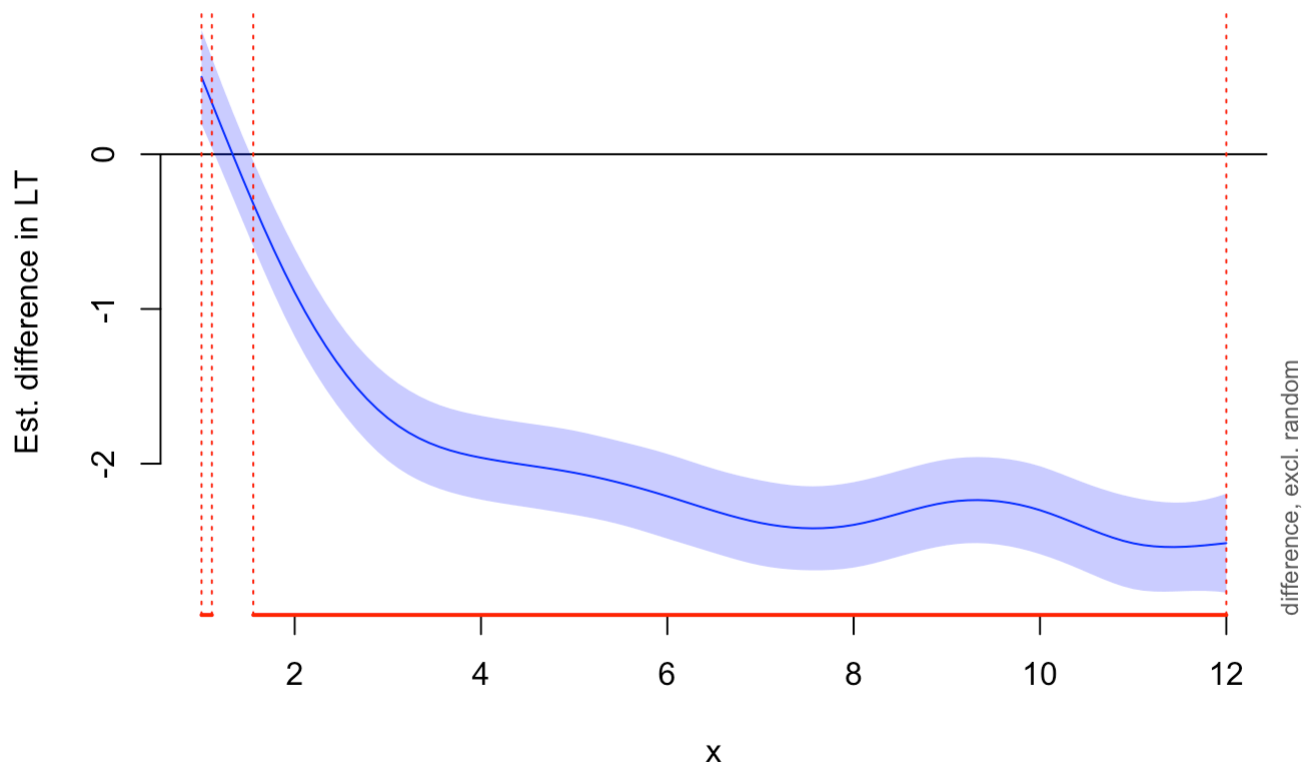


```
##
## x window(s) of significant difference(s):
## 1.000000 - 1.666667
## 2.333333 - 12.000000
```

```
plot_diff(m1.1, view="x",
          comp=list(dir=c(2, 0)),
          # ylim = c(0,8),
          main="ADS-NA",
          col=c('blue','gray'),
          ylab="Est. difference in LT",
          rm.ranef=TRUE)
```

```
## Summary:
## * x : numeric predictor; with 100 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```

ADS-NA

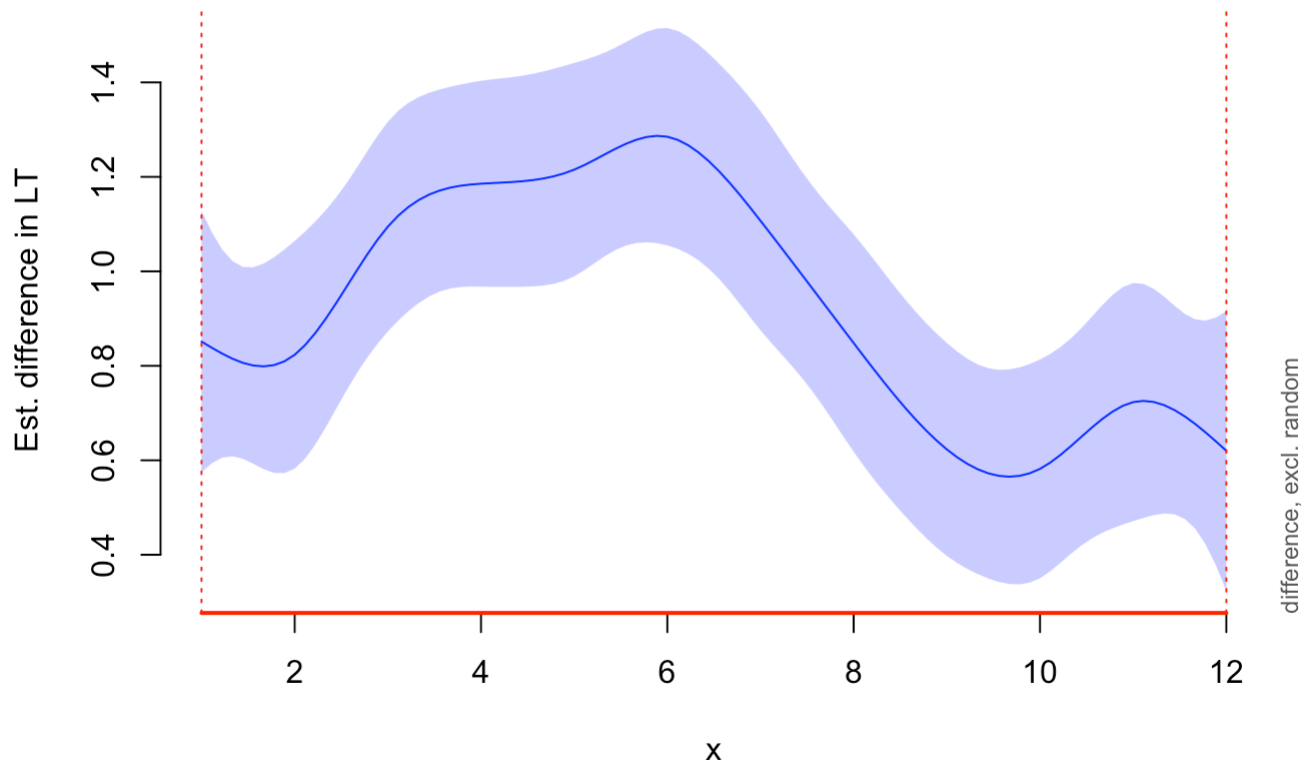


```
##
## x window(s) of significant difference(s):
## 1.000000 - 1.111111
## 1.555556 - 12.000000
```

```
plot_diff(m1.1, view="x",
          comp=list(dir=c(1, 2)),
          # ylim = c(0,8),
          main="IDS-ADS",
          col=c('blue','red'),
          ylab="Est. difference in LT",
          rm.ranef=TRUE)
```

```
## Summary:
## * x : numeric predictor; with 100 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```

IDS-ADS



```
##
## x window(s) of significant difference(s):
## 1.000000 - 12.000000
```

```
# contrasts(sub_df$con) <- 'contr.treatment'
# m1.2 <- bam(y1 ~ s(x) + con + s(x, by = con) + s(id, bs = "re"),
#             data = sub_df,
#             method = "fREML",
#             discrete = TRUE)
# saveRDS(m1.2, file="m1.2.rds")

m1.2 = readRDS("m1.2.rds")

summary(m1.2)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## y1 ~ s(x) + con + s(x, by = con) + s(id, bs = "re")
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.68066    0.06148  124.94  <2e-16 ***
## con1         -1.94291    0.14677  -13.24  <2e-16 ***
## con2         -1.89190    0.14674  -12.89  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf   Ref.df       F p-value
## s(x)           7.496     8.438 446.645  <2e-16 ***
## s(x):con1       3.852     4.741  14.758  <2e-16 ***
## s(x):con2       2.202     2.736  15.056  <2e-16 ***
## s(id)          3460.175 3854.000   9.614  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.517   Deviance explained = 55.8%
## fREML = 1.1243e+05  Scale est. = 11.342    n = 40961
```

```
# par(mfrow=c(2,2), cex=1.1)
```

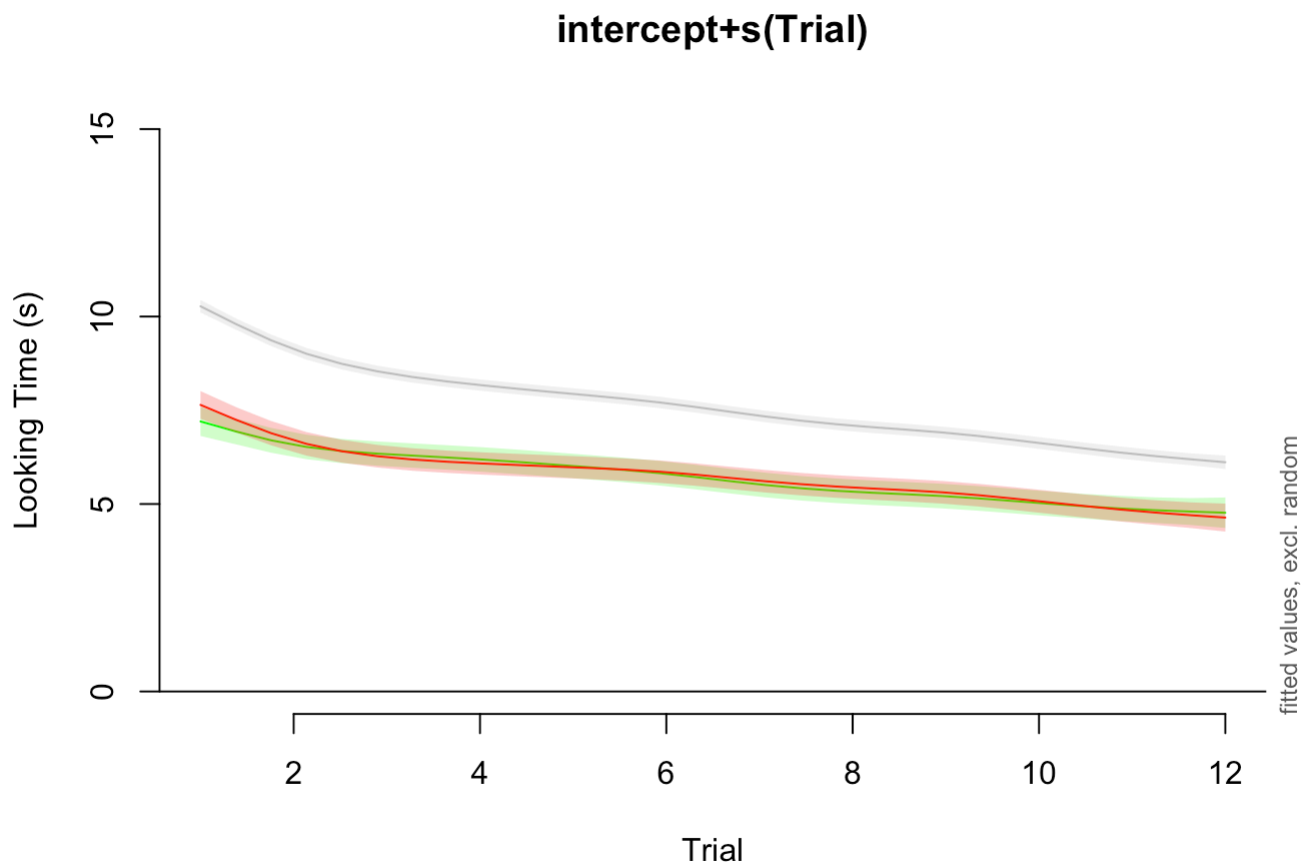
```
plot_smooth(m1.2, view="x", cond=list(con=0),
             xlim = c(1,12), ylim = c(0,15),
             xlab="Trial", ylab="Looking Time (s)",
             rm.ranef=TRUE, main="intercept+s(Trial)",
             col='gray') ## 0 = NA
```

```
## Summary:
## * con : numeric predictor; set to the value(s): 0.
## * x : numeric predictor; with 30 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```

```
plot_smooth(m1.2, view="x", cond=list(con=1),
             add=TRUE, col='green') ## 1 = con
```

```
## Summary:
## * con : numeric predictor; set to the value(s): 1.
## * x : numeric predictor; with 30 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```

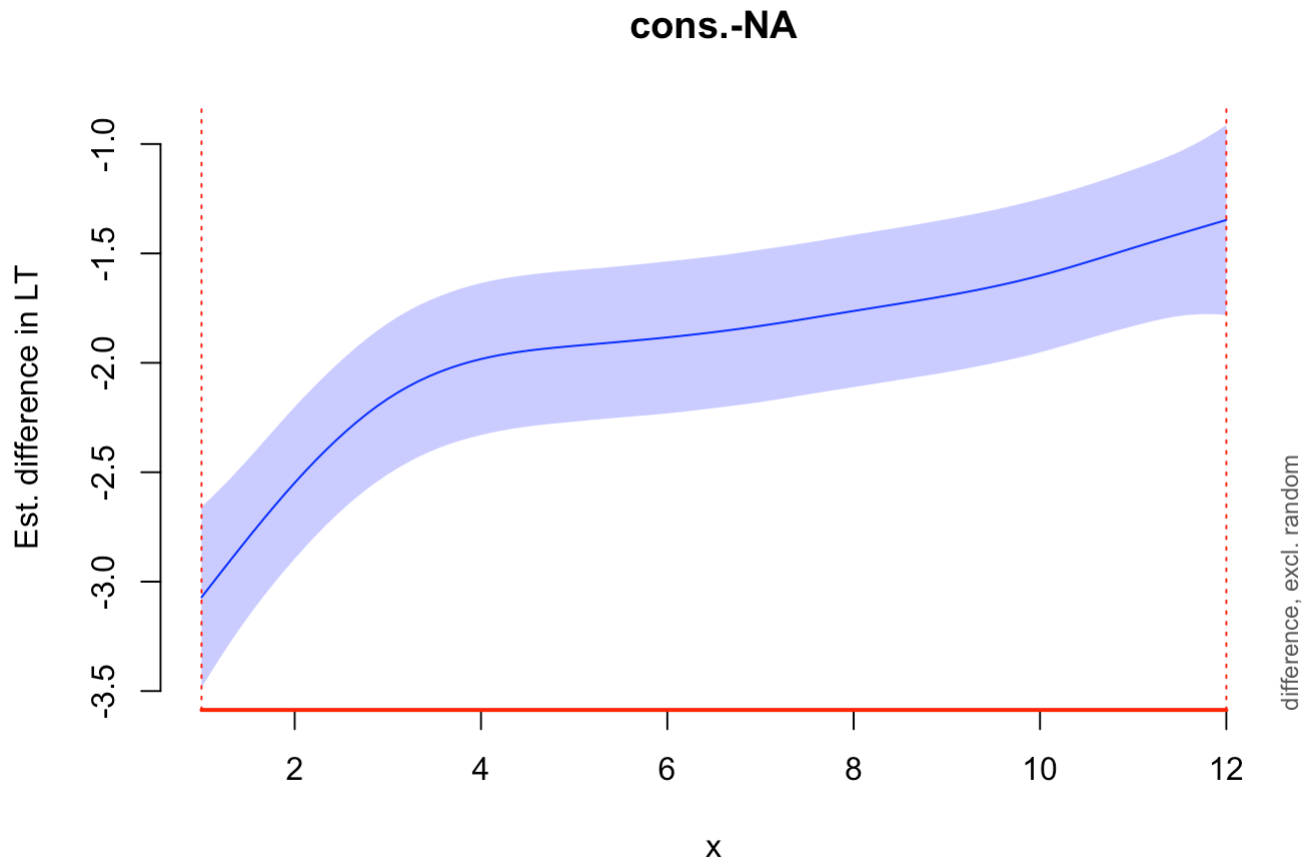
```
plot_smooth(m1.2, view="x", cond=list(con=2),
            add=TRUE, col='red') ## 2 = incon
```



```
## Summary:
## * con : numeric predictor; set to the value(s): 2.
## * x : numeric predictor; with 30 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```

```
plot_diff(m1.2, view="x",
          comp=list(con=c(1, 0)),
          # ylim = c(0,8),
          main="cons.-NA",
          col=c('blue','gray'),
          ylab="Est. difference in LT",
          rm.ranef=TRUE)
```

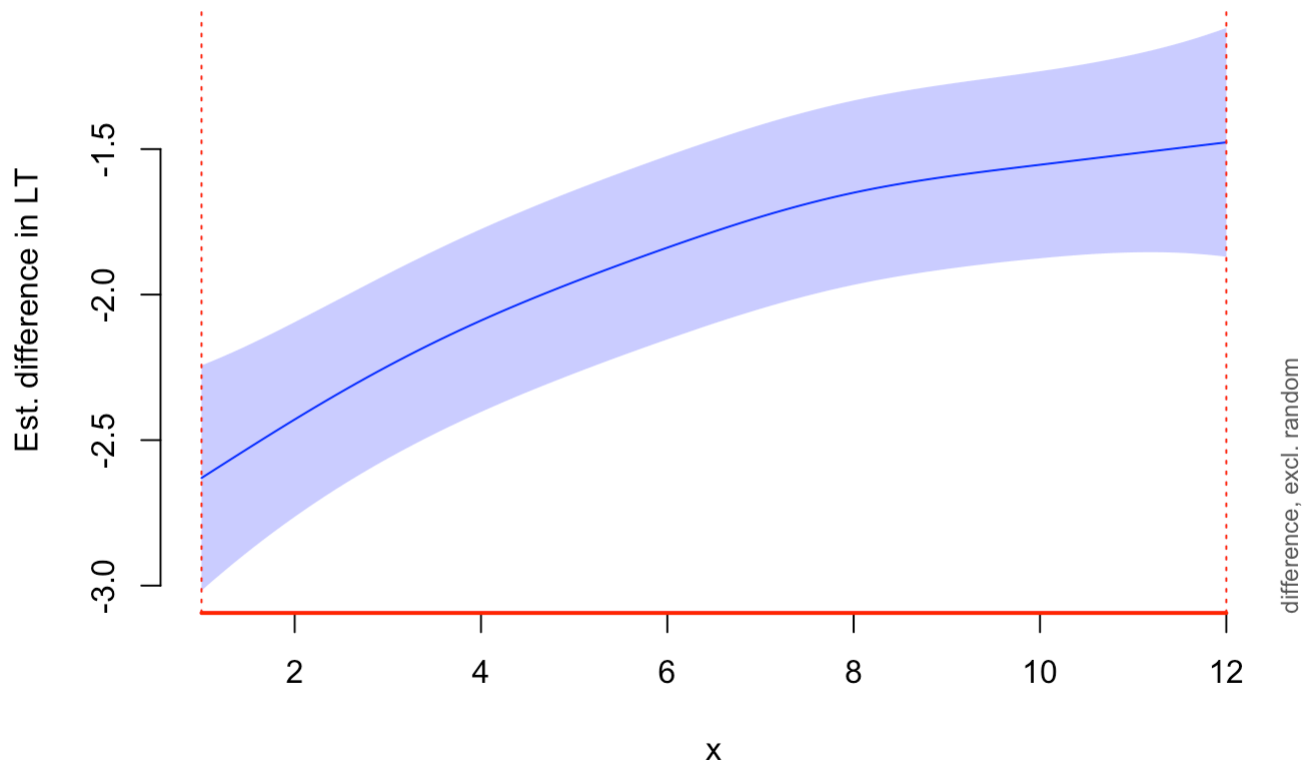
```
## Summary:
## * x : numeric predictor; with 100 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```



```
##
## x window(s) of significant difference(s):
## 1.000000 - 12.000000
```

```
plot_diff(m1.2, view="x",
          comp=list(con=c(2, 0)),
          # ylim = c(0,8),
          main="incon.-NA",
          col=c('blue','gray'),
          ylab="Est. difference in LT",
          rm.ranef=TRUE)
```

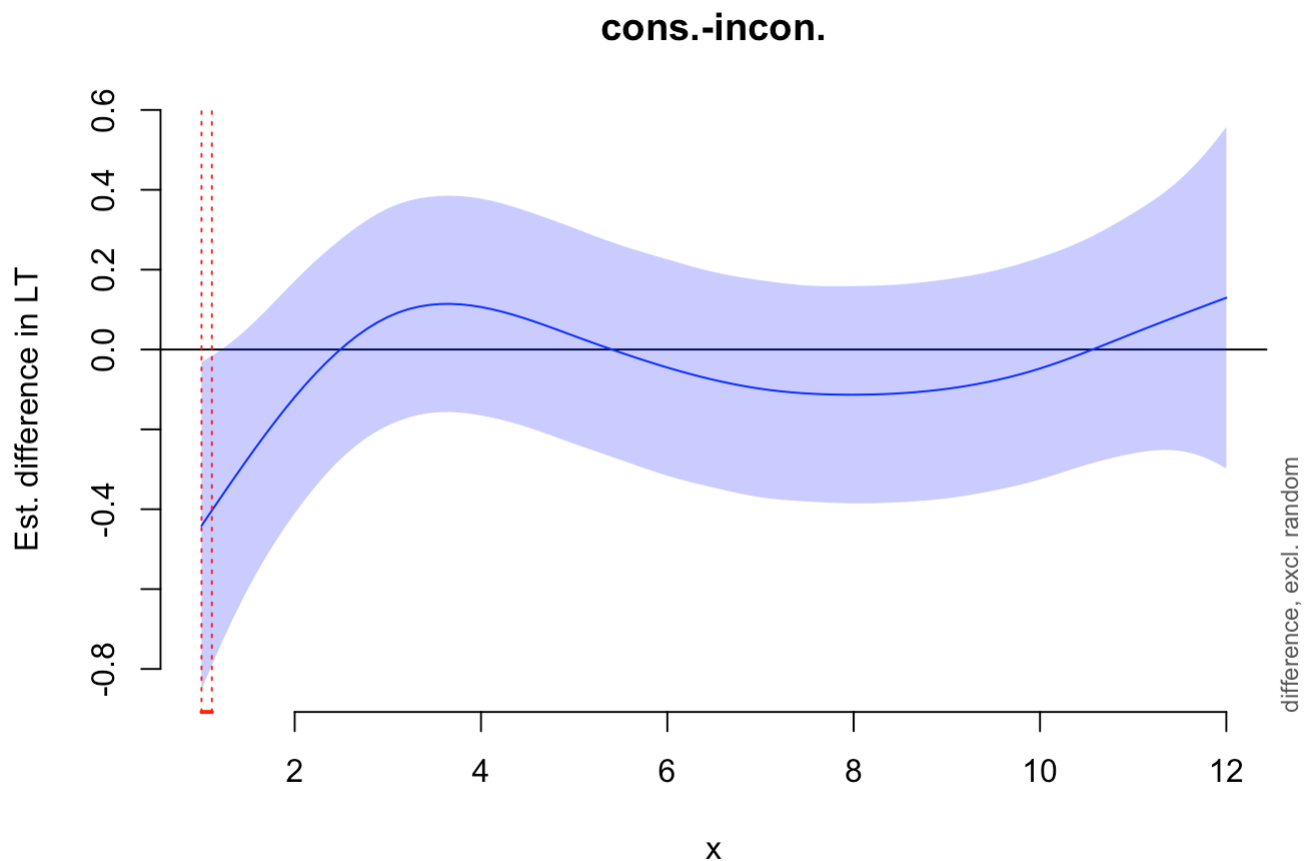
```
## Summary:
## * x : numeric predictor; with 100 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```

incon.-NA

```
##
## x window(s) of significant difference(s):
## 1.000000 - 12.000000
```

```
plot_diff(m1.2, view="x",
          comp=list(con=c(1, 2)),
          # ylim = c(0,8),
          main="cons.-incon.",
          col=c('blue','red'),
          ylab="Est. difference in LT",
          rm.ranef=TRUE)
```

```
## Summary:
## * x : numeric predictor; with 100 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```



```
##
## x window(s) of significant difference(s):
## 1.000000 - 1.111111
```

```
# contrasts(sub_df$soc) <- 'contr.treatment'
# m1.3 <- bam(y1 ~ s(x) + soc + s(x, by = soc) + s(id, bs = "re"),
#             data = sub_df,
#             method = "fREML",
#             discrete = TRUE)
# saveRDS(m1.3, file="m1.3.rds")
```

```
m1.3 = readRDS("m1.3.rds")
```

```
summary(m1.3)
```



```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## y1 ~ s(x) + soc + s(x, by = soc) + s(id, bs = "re")
##
## Parametric coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  6.56076    0.05484  119.63  <2e-16 ***
## soc1         6.02128    0.16417   36.68  <2e-16 ***
## soc2         1.94172    0.16918   11.48  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##           edf   Ref.df       F p-value
## s(x)         7.591     8.495 520.110  <2e-16 ***
## s(x):soc1     4.566     5.597  40.486  <2e-16 ***
## s(x):soc2     3.928     4.851  59.225  <2e-16 ***
## s(id)        3361.484 3853.000    7.593  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.522   Deviance explained = 56.2%
## fREML = 1.1178e+05  Scale est. = 11.226    n = 40961
```

```
# par(mfrow=c(2,2), cex=1.1)
```

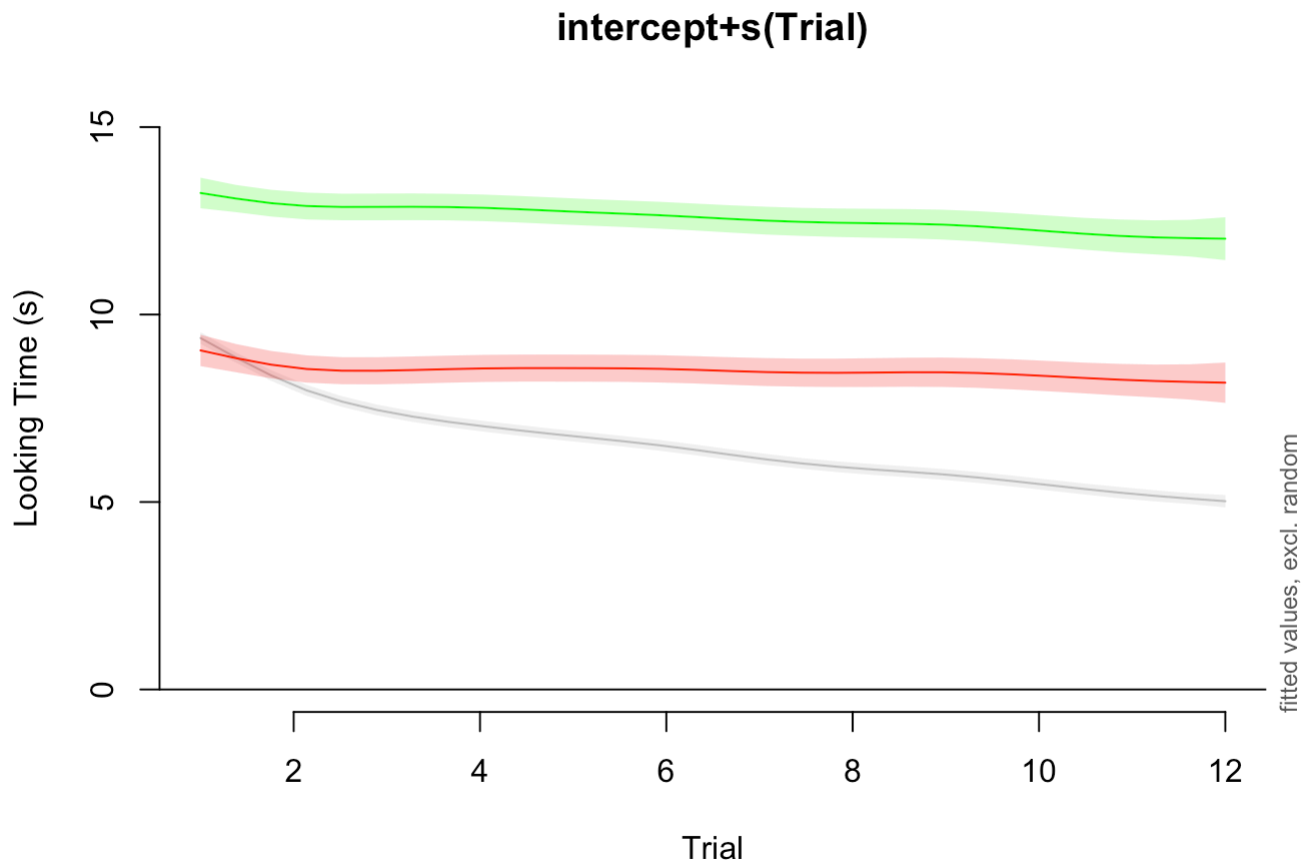
```
plot_smooth(m1.3, view="x", cond=list(soc=0),
            xlim = c(1,12), ylim = c(0,15),
            xlab="Trial", ylab="Looking Time (s)",
            rm.ranef=TRUE, main="intercept+s(Trial)",
            col='gray') ## 0 = NA
```

```
## Summary:
## * soc : numeric predictor; set to the value(s): 0.
## * x : numeric predictor; with 30 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```

```
plot_smooth(m1.3, view="x", cond=list(soc=1),
            add=TRUE, col='green') ## 1 = social
```

```
## Summary:
## * soc : numeric predictor; set to the value(s): 1.
## * x : numeric predictor; with 30 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```

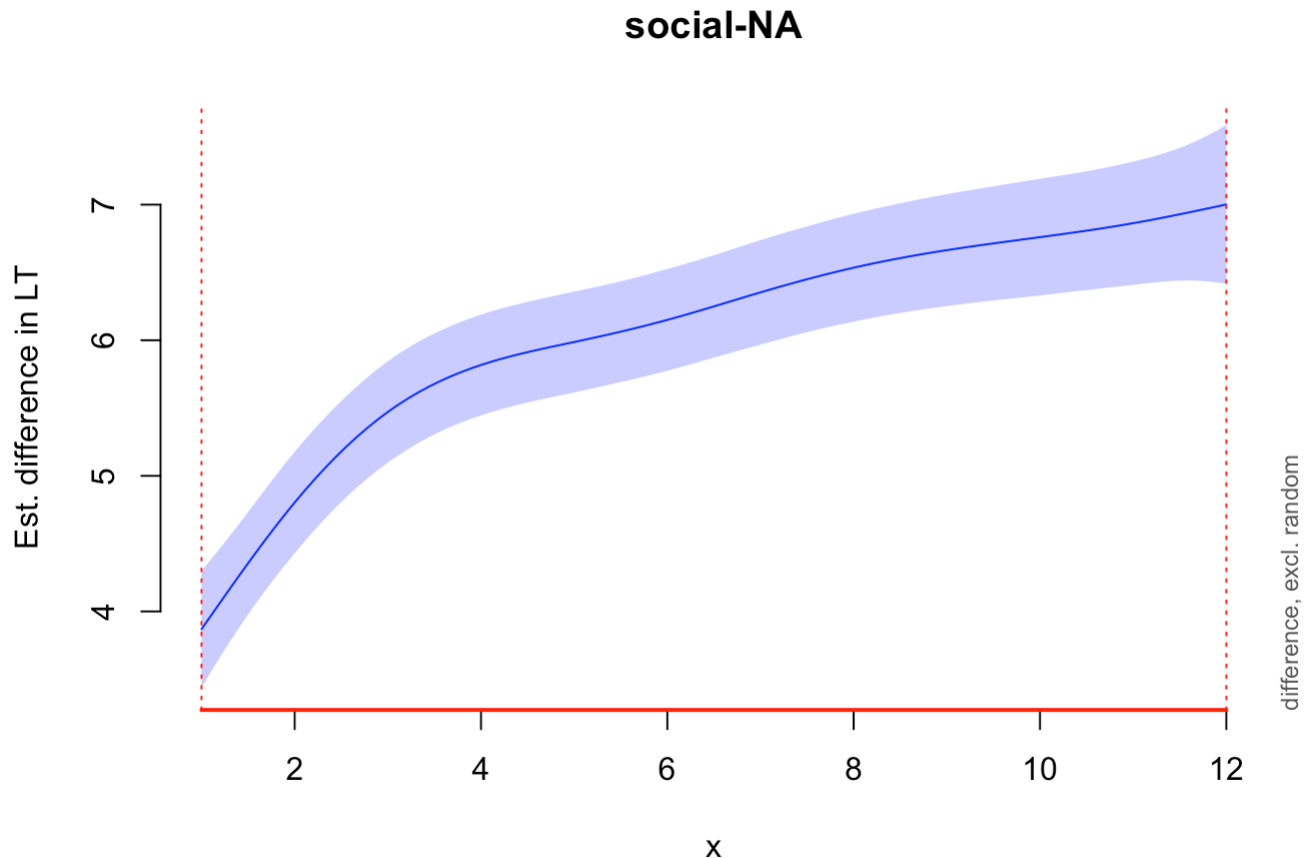
```
plot_smooth(m1.3, view="x", cond=list(soc=2),
            add=TRUE, col='red') ## 2 = nonsocial
```



```
## Summary:
## * soc : numeric predictor; set to the value(s): 2.
## * x : numeric predictor; with 30 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```

```
plot_diff(m1.3, view="x",
          comp=list(soc=c(1, 0)),
          # ylim = c(0,8),
          main="social-NA",
          col=c('blue','gray'),
          ylab="Est. difference in LT",
          rm.ranef=TRUE)
```

```
## Summary:
## * x : numeric predictor; with 100 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```

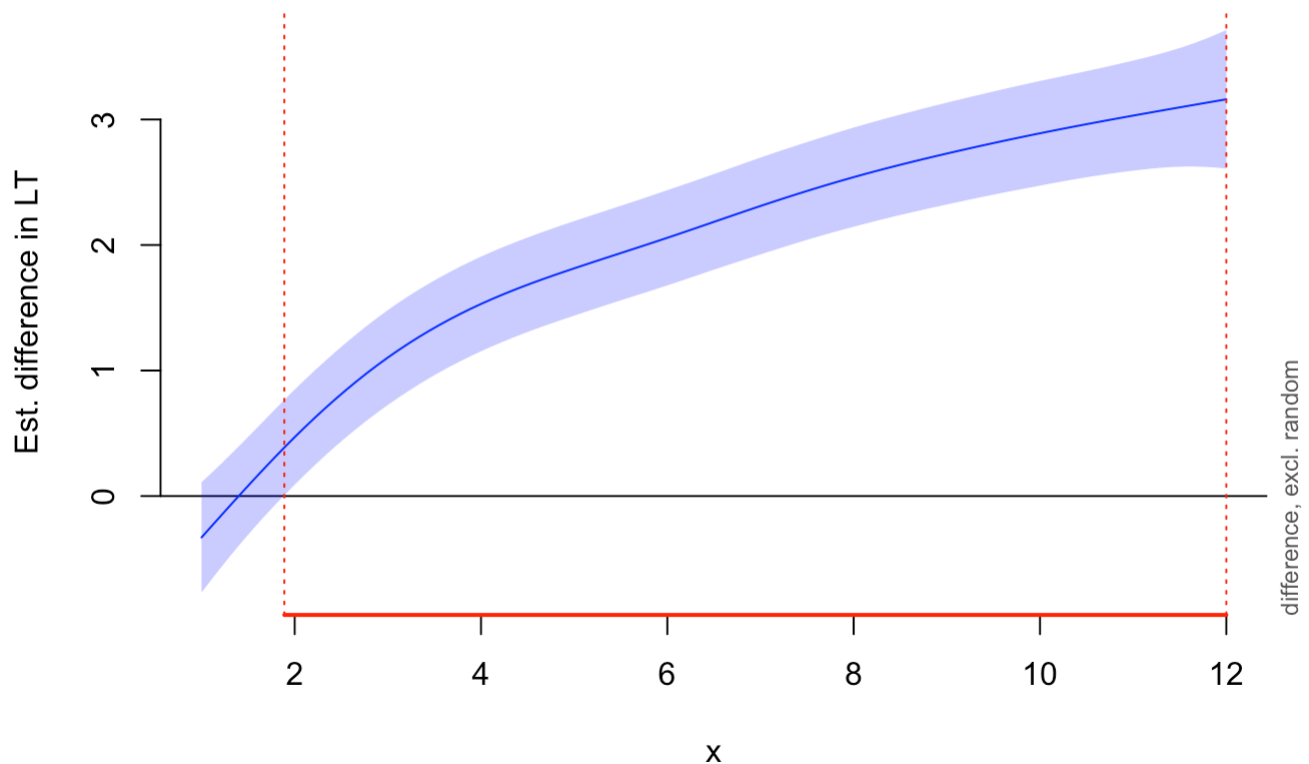


```
##
## x window(s) of significant difference(s):
## 1.000000 - 12.000000
```

```
plot_diff(m1.3, view="x",
          comp=list(soc=c(2, 0)),
          # ylim = c(0,8),
          main="nonsocial-NA",
          col=c('blue','gray'),
          ylab="Est. difference in LT",
          rm.ranef=TRUE)
```

```
## Summary:
## * x : numeric predictor; with 100 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookes_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```

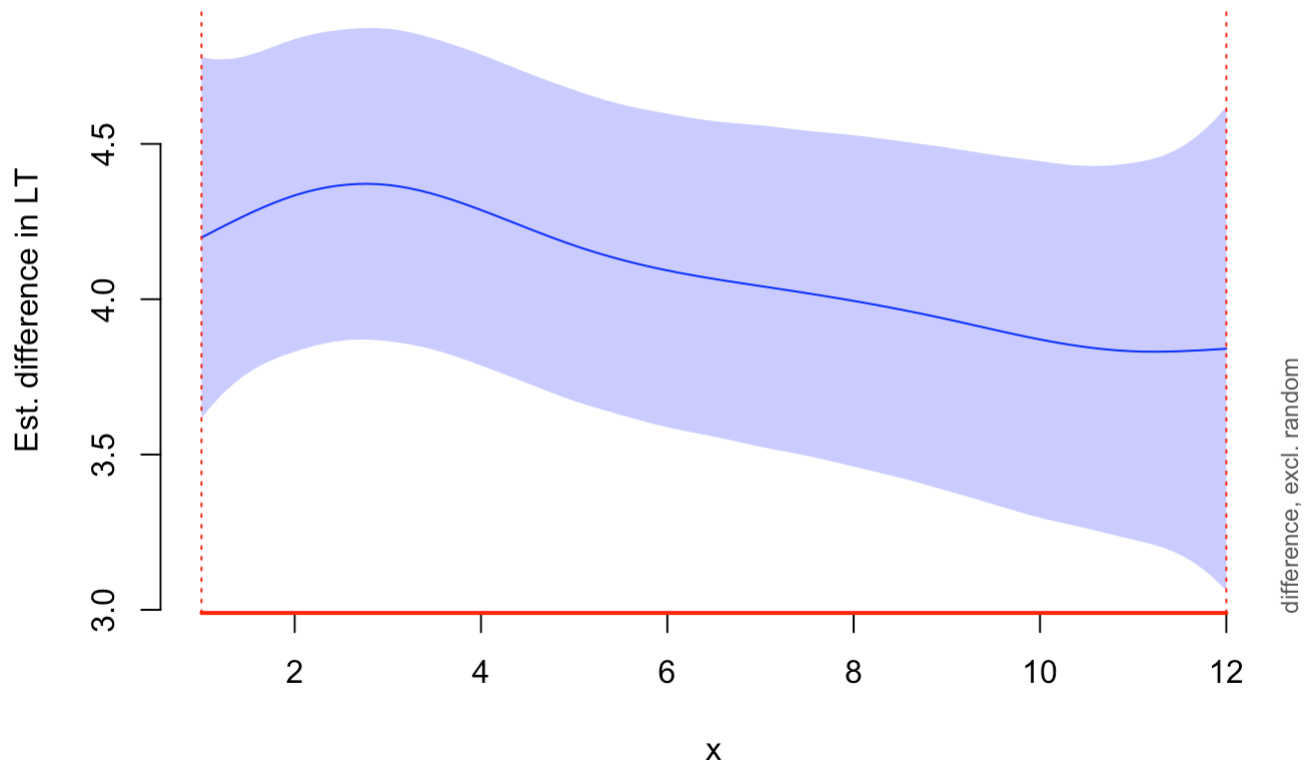
nonsocial-NA



```
##
## x window(s) of significant difference(s):
## 1.888889 - 12.000000
```

```
plot_diff(m1.3, view="x",
          comp=list(soc=c(1, 2)),
          # ylim = c(0,8),
          main="social-nonsoc",
          col=c('blue','red'),
          ylab="Est. difference in LT",
          rm.ranef=TRUE)
```

```
## Summary:
## * x : numeric predictor; with 100 values ranging from 1.000000 to 12.000000.
## * id : factor; set to the value(s): MB1_babylabbrookess_mb01. (Might be canceled a
s random effect, check below.)
## * NOTE : The following random effects columns are canceled: s(id)
##
```

social-nonsoc

```
##  
## x window(s) of significant difference(s):  
## 1.000000 - 12.000000
```